

KARLSBORG, Sweden

WMO: 025440

Lat: 58.517N

Lon: 14.517E

Elev: 94

StdP: 100.20

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.6	-15.0	1.0	-12.4	-12.4	1.3	-9.5	11.9	4.5	10.5	3.6	2.5	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	25.7	18.5	24.0	17.6	22.1	17.0	19.9	23.6	18.9	22.4	17.9	20.9	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	21.4	17.6	12.7	20.3	16.6	11.9	19.3	57.3	23.8	53.9	22.4	50.7	21.0	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.2	9.0	8.0	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-14.9	20.7	5.1	1.3	-18.5	21.6	-21.5	22.4	-24.4	23.1	-28.1	24.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-1.8	-1.4	1.3	5.8	10.7	14.5	17.1	15.5	12.2	7.5	3.9	0.3	(6)
(7)		DBStd	7.38	4.30	4.58	3.86	3.14	3.55	2.62	2.73	2.39	2.53	3.02	3.41	4.59	(7)
(8)		HDD10.0	1701	367	320	269	129	33	1	0	0	8	89	185	300	(8)
(9)		HDD18.3	4100	625	553	527	375	236	119	58	93	184	337	434	558	(9)
(10)		CDD10.0	674	0	0	1	4	56	134	221	172	74	11	1	0	(10)
(11)		CDD18.3	30	0	0	0	0	0	3	21	6	0	0	0	0	(11)
(12)		CDH23.3	222	0	0	0	1	8	38	147	29	0	0	0	0	(12)
(13)		CDH26.7	28	0	0	0	0	0	2	22	4	0	0	0	0	(13)
(14)	Wind	WSAvg	3.8	4.0	4.0	4.0	3.5	3.4	3.4	3.2	3.3	3.7	4.2	4.4	4.2	(14)
(15)	Precipitation	PrecAvg	711	48	42	32	42	54	85	82	78	64	64	60	53	(15)
(16)		PrecMax	876	92	81	57	102	111	155	173	129	139	135	118	108	(16)
(17)		PrecMin	550	10	8	5	9	16	4	20	15	10	8	18	16	(17)
(18)		PrecStd	93	22	18	13	22	24	36	36	33	40	30	27	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	9.0	13.8	18.2	24.5	26.5	29.2	26.0	21.4	15.7	11.7	10.4	(19)
(20)			MCWB	6.5	6.4	9.4	10.6	15.4	17.4	19.4	18.9	17.3	13.6	11.1	9.7	(20)
(21)		2%	DB	5.6	6.7	10.4	15.9	22.2	24.3	26.6	23.5	19.1	14.3	10.6	9.3	(21)
(22)			MCWB	4.7	4.7	7.0	10.0	14.3	16.7	19.1	18.3	15.9	12.6	10.0	8.5	(22)
(23)		5%	DB	4.6	5.5	8.3	13.9	19.7	22.2	24.8	21.5	17.7	13.2	9.4	7.4	(23)
(24)			MCWB	3.7	4.3	5.6	9.0	14.0	15.9	18.5	17.3	14.8	11.8	8.8	6.5	(24)
(25)		10%	DB	3.4	4.5	6.6	11.7	17.4	20.2	22.9	20.1	16.3	12.1	8.4	5.8	(25)
(26)			MCWB	2.6	3.5	4.6	7.6	12.5	15.0	18.1	16.7	14.0	10.8	7.7	4.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	6.8	9.6	12.5	17.2	19.1	21.4	20.7	17.7	14.2	11.4	10.0	(27)
(28)			MCDWB	7.1	8.0	13.2	16.9	21.7	23.2	25.2	24.0	20.4	15.1	11.7	10.2	(28)
(29)		2%	WB	4.9	5.3	7.5	10.5	15.9	17.8	20.6	19.1	16.4	13.1	10.1	8.3	(29)
(30)			MCDWB	5.5	6.2	9.9	14.4	20.1	22.6	24.4	21.8	18.5	14.1	10.6	9.1	(30)
(31)		5%	WB	3.7	4.5	6.0	9.4	14.4	16.7	19.6	18.1	15.4	12.0	8.9	6.6	(31)
(32)			MCDWB	4.4	5.4	7.8	13.0	18.4	21.0	23.5	20.5	17.0	12.8	9.3	7.3	(32)
(33)		10%	WB	2.7	3.5	4.8	8.1	13.1	15.6	18.6	17.3	14.4	11.0	7.9	4.9	(33)
(34)			MCDWB	3.4	4.4	6.3	11.5	16.9	19.5	22.3	19.5	15.8	11.9	8.3	5.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.3	4.8	6.3	8.8	9.4	9.1	8.5	7.9	6.8	5.3	3.9	4.2	(35)
(36)			MCDBR	5.1	5.7	8.3	12.5	13.1	12.5	11.7	10.1	8.7	6.0	4.7	5.1	(36)
(37)			MCWBR	4.9	4.4	5.9	7.4	7.2	6.7	5.5	5.8	5.7	4.7	4.5	5.2	(37)
(38)		5% WB	MCDWB	5.1	5.1	7.9	10.9	11.3	11.4	10.2	8.9	8.2	5.6	4.5	5.1	(38)
(39)			MCWBR	4.8	4.1	6.0	6.9	6.9	6.8	5.6	5.6	5.8	4.8	4.3	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.313	0.334	0.372	0.369	0.358	0.381	0.373	0.343	0.333	0.302	0.261	(40)	
(41)		taud	2.228	2.293	2.359	2.326	2.398	2.449	2.403	2.422	2.493	2.435	2.303	2.165	(41)	
(42)		Ebn at Noon	557	713	811	833	864	879	847	826	799	688	521	440	(42)	
(43)		Edh at Noon	46	70	87	107	106	103	105	96	77	62	43	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.35	0.96	2.48	4.00	5.16	5.77	5.16	4.09	2.77	1.26	0.45	0.24	(44)	
(45)		RadStd	0.05	0.16	0.28	0.45	0.45	0.27	0.61	0.43	0.38	0.21	0.08	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-128	-159	N/A	N/A

Nomenclature: See separate page